

EduGenie: Google Gemini Powered Learning Assistant

Phase 5

3. Quiz Module

EduGenie generates three multiple-choice questions (MCQs) from a given passage, each containing four carefully crafted options. It utilizes the Gemini model to comprehend the context and meaning of the passage, ensuring that the questions are relevant and the distractors are plausible. The final output is structured in JSON format for easy integration.

Fig. Quiz Module Logic

It sends a structured prompt to the model, expecting a valid JSON response with each question containing four options and a correct answer. To ensure proper parsing, it cleans any Markdown code blocks using the `clean_json_blockfunction`. The cleaned response is then parsed into a Python list. If an error occurs during generation or parsing, it returns a detailed error message for debugging.

4. Summary Module

This feature leverages Gemini's generative capabilities to summarize long paragraphs into concise, easy-to-understand versions, making it ideal for quick revision. It ensures the core information is retained while eliminating redundancy, helping learners absorb key points efficiently without losing clarity or context.

Fig. Summary Module